

1.2 - Gap Analysis :

Three requirements with no existing HW1-HW4 coverage were identified. A new pytest test was written for each gap in tests/test_hw5_rtm.py.

Requirement	Why it was not tested in HW1-HW4	New Test Case
R-APPT-LATE: Late cancellation (<24 hours) must set is_late_cancellation=True.	This is a time-based edge case. Earlier assignments mainly tested normal appointment scheduling and cancellation, not cancellations within the 24-hour window.	test_r_appt_late_cancellation_sets_late_flag()
R-BILL-VOID: Cannot process payment on a Voided invoice.	Previous assignments focused on successful billing and payment workflows. Invalid invoice states such as Voided were not exercised.	test_r_bill_void_payment_not_allowed()
R-REC-AUDIT: Medical record updates should preserve an audit trail.	Audit logging is a secondary requirement that is easy to overlook when functional behavior is the primary testing focus.	test_r_rec_audit_trail()

Security Testing Report

2.1 - SQL Injection Testing

Functions tested: search_patients() and get_patient(). Five common SQL injection payloads were executed against each function. Each test verified that the application did not crash, the patients table still existed, and the payload was not reflected in the returned results.

Result: 10/10 SQL injection tests is passed

Task 2.2 - Authentication and Authorization Testing

Security Test	Result
Inactive patient cannot schedule appointment	Passed
Wrong patient cannot add diagnosis for another patient's appointment	Passed
Cancelled appointment cannot receive diagnosis	Passed
Invalid payment method rejected	Passed
Zero payment without insurance rejected	Passed

Overall Results

SQL Injection Tests: 10 Passed, 0 Failed

Authentication & Authorization Tests: 5 Passed, 0

Failed Total Security Tests: 15 Passed, 0 Failed.

Conclusion:

The MediTrack application successfully handled the SQL injection payloads used in this assignment and correctly enforced the security-related business rules covered by the implemented tests. All security tests completed successfully.